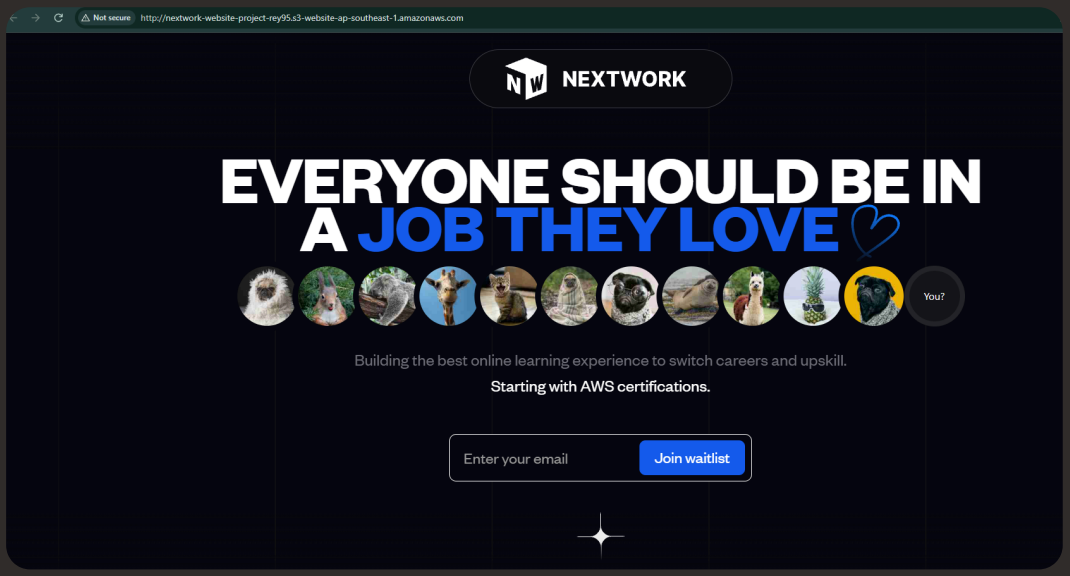
HOST A WEBSITE ON AMAZON S3

By Reynaldo Eusebio | September 2026



Introducing Today's Project!

Project overview

In this project, I will demonstrate how to host a static website using Amazon S3. I am doing this project to learn how to create an S3 bucket, upload website files, and configure static website hosting.

Tools and concepts

Services I used were Amazon S3. Key concepts I learned include S3 buckets, static website hosting, ACLs, and bucket policies. I learned how to create an S3 bucket, host a static website on it, and control access to website files using ACLs and bucket policies.

Time, challenges, and wins

This project took me approximately 45 minutes. The most challenging part was understanding how to configure the permissions for my website files. It was most rewarding to successfully host the static website and see it working online.

How I Set Up an S3 Bucket

What I did in this step

In this step, I created a bucket in Amazon S3 because I will use it to store my static website files.

How long it took to create the bucket

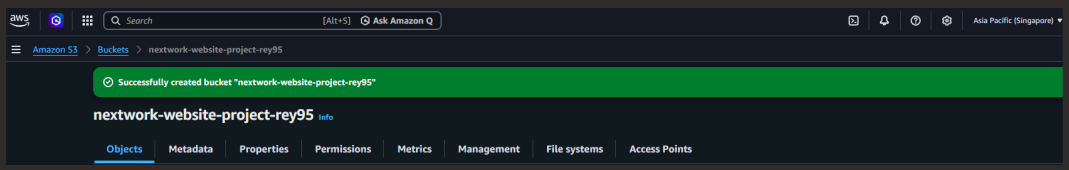
Creating an S3 bucket took me less than an hour.

Region selection

I selected the Singapore (ap-southeast-1) Region because it is geographically close to the Philippines, which can help reduce latency.

Understanding bucket name uniqueness

S3 bucket names must be globally unique within the AWS partition, which means no other bucket can use the same name.



Upload Website Files to S3

What I did in this step

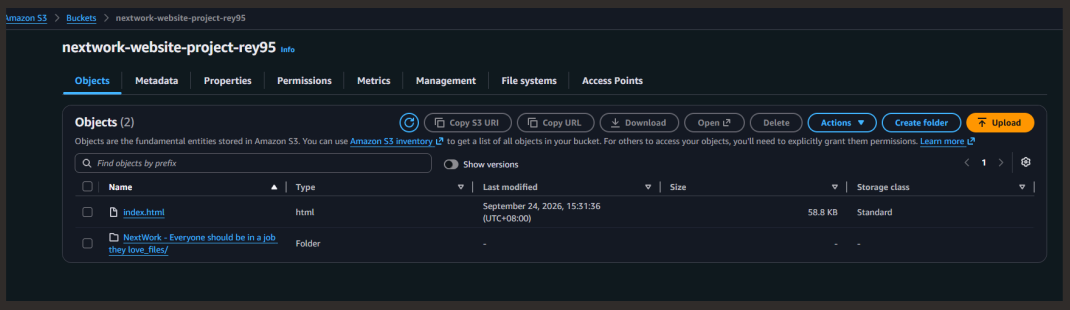
In this step, I uploaded the website files because they need to be stored in the S3 bucket before the website can be hosted.

Files I uploaded

I uploaded `index.html` and the folder containing the website's image assets to my S3 bucket.

How the files work together

Both files are necessary for this project as the `index.html` file contains the structure and content of the website, while the image assets provide the images and other resources that the HTML file uses to display the complete website.



Static Website Hosting on S3

What I did in this step

In this step, I will configure static website hosting on my S3 bucket because I need to make my website accessible on the internet.

Understanding website hosting

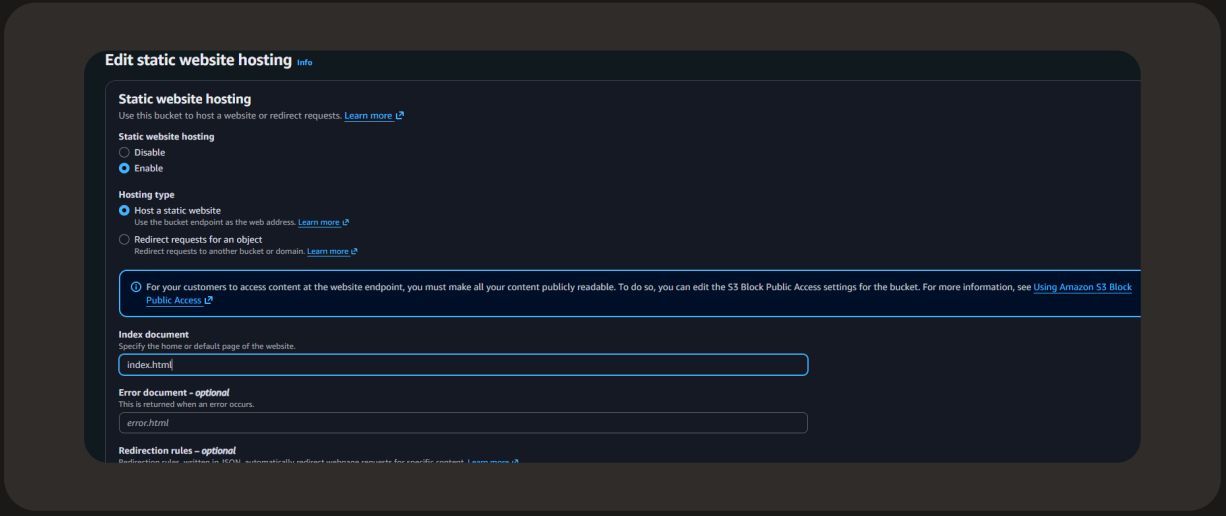
Website hosting means putting your website files on a special computer called a web server, where they can be made accessible to other people anytime and anywhere as long as they have an internet connection.

How I enabled website hosting

To enable website hosting with my S3 bucket, I uploaded my website's files to the S3 bucket and enabled static website hosting in the Properties settings.

Access Control Lists (ACLs)

An ACL is used to control who can access an object. I enabled it because it is useful for controlling access to the website files in this project.



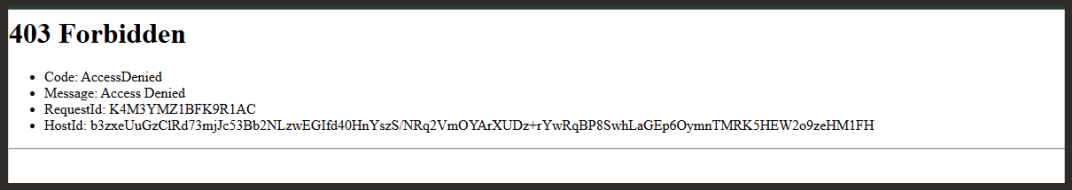
Bucket Endpoints

Understanding bucket endpoint URLs

Once static website hosting is enabled, S3 provides a bucket endpoint URL, which is my website's web address.

What I saw when I tested the endpoint

When I first visited the bucket endpoint URL, I saw a 403 Forbidden error. The reason for this error was that my website files were still private, so the public did not have permission to access them.



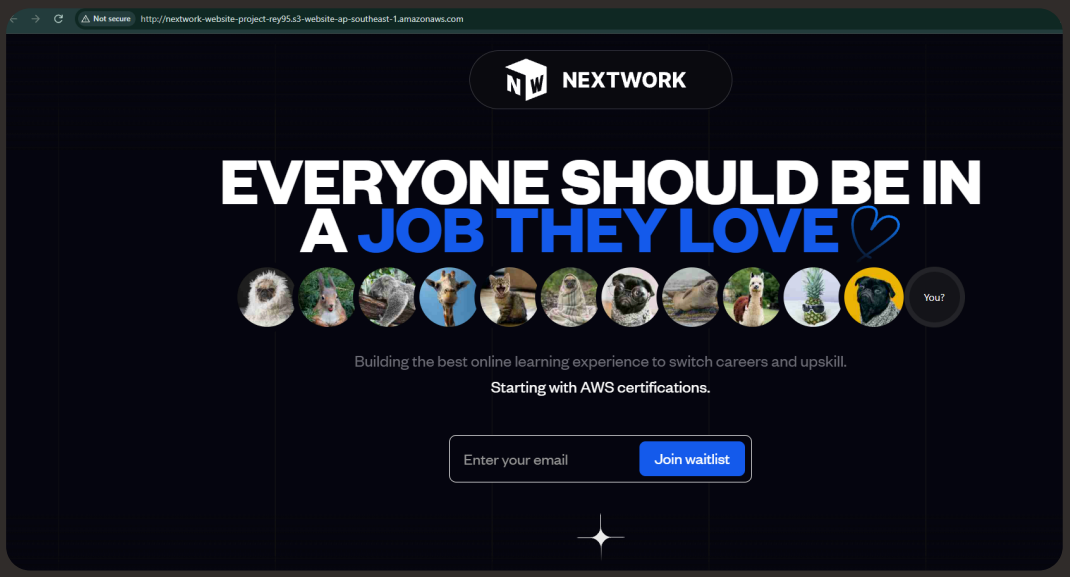
Success!

What I did in this step

In this step, I will make my website files publicly accessible because people need permission to view them when accessing my website.

How I resolved the 403 error

To resolve this 403 Forbidden error, I made my website files public using ACLs.



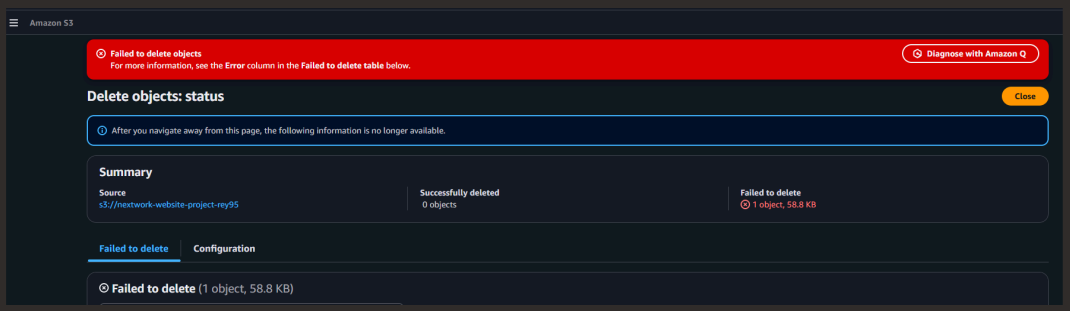
Bucket Policies

What I did in this extension

In this project extension, I'm about to create a bucket policy to prevent people from deleting my index.html file. I'm doing this so that the file cannot be deleted by others.

Understanding bucket policies

An alternative to ACLs are bucket policies, which are rules written in JSON that control access to resources in an S3 bucket. The benefit of using bucket policies is that they provide more specific control over actions and permissions, while ACLs are useful for controlling access to individual objects.



What my bucket policy does

My bucket policy prevents the deletion of my index.html file. I tested this by trying to delete the file and saw that the deletion was denied.